Department of Computer Science and Engineering

Course Code:CSE430 **Credits: 1.5**

Course Name: Digital Signal Processing (DSP) **Prerequisite:CSE111, MAT120**

Lab 04

Up and Down Sampling, LTI System, and Convolution

I. Lab Overview:

Learn MATLAB programming concepts for Up and Down Sampling, LTI System, and Convolution.

II. Up/Down Sampling, System Response, and Convolution:

Up Sampling:

The up-sampler is a discrete-time system defined by the input-output relation

$$y(n) = \begin{cases} x(\frac{n}{L}); & n = 0, \pm L, \pm 2L, \pm 3L \text{ etc..} \\ 0, & \text{otherwise} \end{cases}$$

L is called the up-sampling factor which is a positive integer greater than 1. L-1 equidistant zero samples are inserted by the up-sampler between two input samples. The effect of up sampling is to expand the signal in time domain without discarding information from the main signal. It finds applications in sampling rate alteration process.

Down Sampling:

In down sampling the pre-sampled signal is again sampled at some rate. The down sampling operation is mathematically represented as:

y(n) = x(nL); where L is an integer (Down sampling rate)

The effect of down sampling is to shrink the signal in time domain at the cost of discarding some information from the original signal.

Convolution

Any arbitrary input signal can be decomposed and represented as a weighted sum of unit sample sequences. As a consequence of linearity and time-invariance properties of the system, the response of the system to any arbitrary input signal can be expressed in terms of the unit sample response of the system. The general form of the expression that relates the unit sample response of the system and arbitrary input signal to the output signal is called the convolution sum.

For a relaxed LTI system, the response y(n) to a given input signal x(n) can be obtained if we know the system impulse response/unit sample response h(n) . Then response y (n) is given by the following relation

$$y(n) = \sum_{k=-\infty}^{k=\infty} x(k)h(n-k)$$

which states that y = convolution(x,h)

Steps to find the convolution are

Step1: Find the index of the result. Starting index of the result is obtained by summing the first indices of x and h.

Step 2: Fold the signal h(k) to generate h(-k).

Step 3: Shift the folded signal h(-k) by n samples to generate h(n-k).

Step 4: Multiply elements of x(k) with corresponding elements from x(n-k) by matching the indices.

Step 5: Sum the product terms generated in Step 4.

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System Response from difference equations

An important subclass of LTI discrete time (DT) systems is characterized by a linear constant co-efficient difference equation of the form

$$\sum_{k=0}^{N} a_k y(n-k) = \sum_{k=0}^{M} b_k x(n-k)$$

where x(n) and y(n) are, respectively, the input and output of the system. (N≥M assumed).

$\{a_k\}$ and $\{b_k\}$ are the coefficients. We can modify the equation as:

$$y(n) = -\sum_{k=1}^{N} \frac{a_k}{a_0}\ y(n-k) + \sum_{k=0}^{M} \frac{b_k}{a_0}\ x(n-k)$$

if, $a_0 = 1$

$$y(n) = -\sum_{k=1}^{N} a_k\ y(n-k) + \sum_{k=0}^{M} b_k\ x(n-k)$$

Based on the above equation, a causal LTI system can be simulated in MATLAB using filter( ) function. Its general form is:

y = filter(b, a ,x); ; remember co-efficient a_0 must be non-zero.

If a_0 is not equal to 1, *filter( )* normalizes the system coefficients by a_0

III. Lesson Fit:

There is pre-requisite to this lab: CSE111, MAT120. You should have intensive Programming Knowledge and capability of understand Integral Calculus and Differential Equations.

IV. Acceptance and Evaluation

Performed lab tasks will be evaluated by the Lab Instructor (LI)

a. Short viva will be conducted in each Lab or occasionally to examine your work.
b. You may work in groups but be aware that you will be evaluated individually; hence active participation during the Lab work demonstration is recommended.
c. There will be Lab handout after your work you have to handover it to LI

V. Learning Outcome:

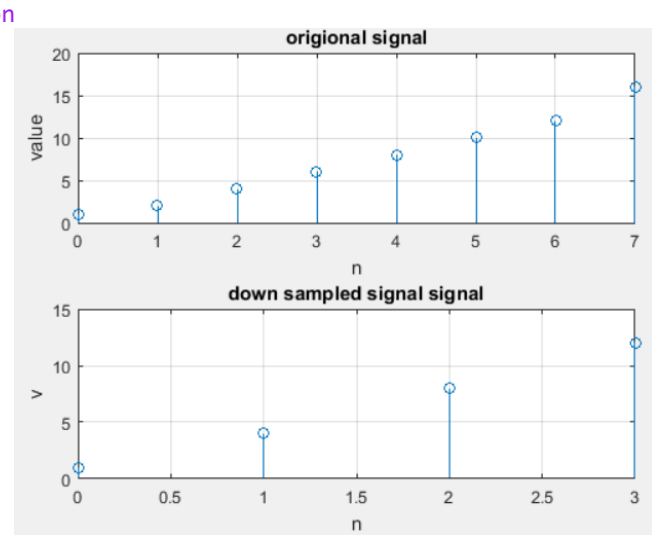
After this Lab, the students will be able to:

a. Understand basic of elementary discrete-time signals.

VI. Activity Detail

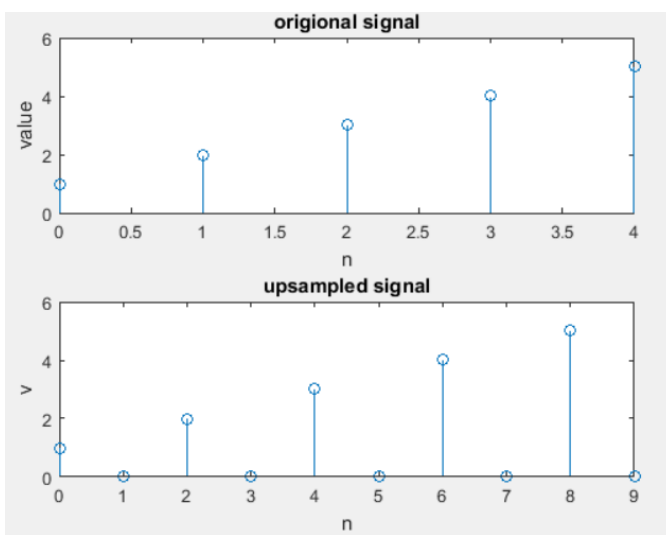
a. Down Sampling (Hour: 1.50)

```
x=[1 2 4 6 8 10 12 16]
n=[0: length(x)-1]
factor=input('samping by factor:')
x_downsampled=[ ]
for i=1:factor:length(x)
    x_downsampled=[x_downsampled x(i)]
end
grid on
subplot 211
stem(n, x)
xlabel('n')
ylabel('value')
title('origional signal')
grid on
subplot 212
stem(0:length(x_downsampled)-1,x_downsampled)
xlabel('n')
ylabel('v')
title('down sampled signal signal')
grid on
```



b. Up Sampling (Hour: 1.50)

```
x=[1 2 3 4 5]
n=[0: length(x)-1];
factor=input('samping by factor:');
x_upsampled=[];
for i=1:length(x)
    x_upsampled=[x_upsampled x(i)]
    for j=1:factor-1
       x_upsampled=[x_upsampled 0]
    end
end
subplot 211
stem(n, x)
xlabel('n')
ylabel('value')
title('origional signal')
subplot 212
stem(0:length(x_upsampled)-1,x_upsampled)
xlabel('n')
ylabel('v')
title('upsampled signal')
```



c. Convolution (Hour: 1.50)

```
clc;
clear all;
```

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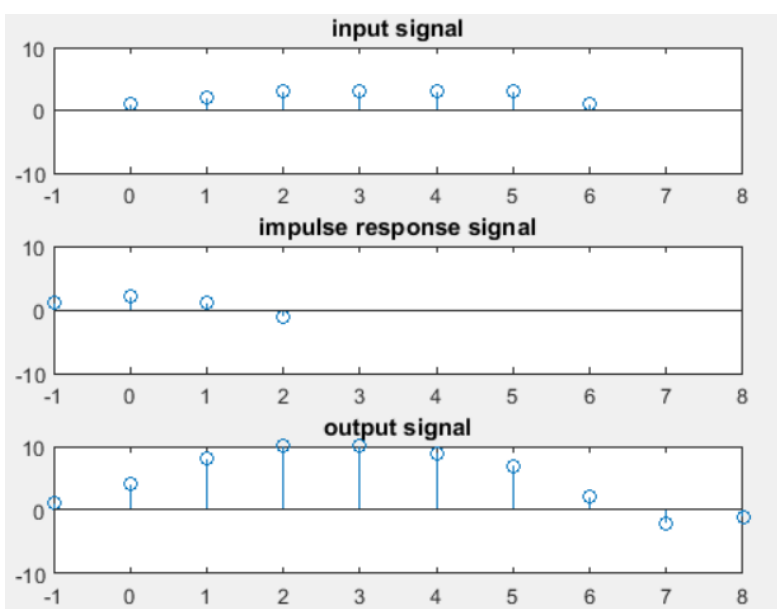
```
close all;

h=[1 2 1 -1];  %values of h
h_id=[-1:2];   %index of h
x=[1 2 3 3 3 3 1];   %values of x
x_id=[0:6];    %index of x

idmin=h_id(1)+x_id(1);
idmax=h_id(end)+x_id(end);
id_y=idmin:idmax;  %generating index of the
result/convolution output

for n=1:length(id_y)
    [h1_id,h1]=sigfold(h_id,h);  %folding the signal
    h1_id=h1_id+id_y(n);   %shifting the folded signal
    p=zeros(1,length(h1_id));

    for k=1:length(h1_id)    % for doing sum of x(k)*h(n-k)
        i=h1_id(k);          %finding index h1
        i2=find(x_id==i);    %matching the corresponding
index from x
        if (i2)
            p(k)=h1(k)*x(i2);
        end
    end
    y(n)=sum(p);             %result/convolution sum
end
subplot(311),stem(x_id,x),title('input
signal'),axis([idmin idmax -10 10])
subplot(312),stem(h_id,h),title('impulse response
signal'),axis([idmin idmax -10 10])
subplot(313),stem(id_y,y),title('output
signal'),axis([idmin idmax -10 10])
```



The function conv( ) only gives the values of the convolution, it does not provide information about the indices. If you need the corresponding indices of the convolution, you must define a function that gives the indices as well as the result of the convolution sum.

d. Impulse response, step response and sinusoidal response (Hour: 1.50)

Consider a system described by the following equation

$y(n) + 0.6y(n-1) = x(n)$

Use MATLAB to find impulse response, step response and sinusoidal response for

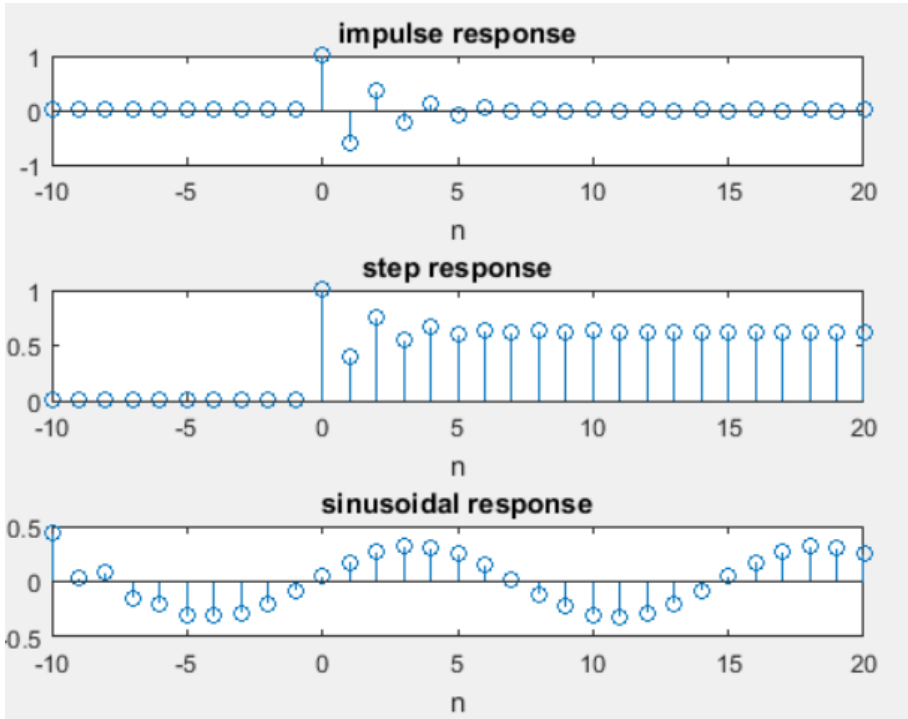
$x(n)=0.5\sin(\omega n)u(n)$ in the range $-10 \leq n \leq 20$.

```
clc
close all
a=[1, 0.6];
b=[1];
n=-10:20; %generating index
figure,
%Finding Impulse Response
x=zeros(1,length(n));
ir=find(n==0);
x(ir)=1; %generating impulse sequence
y=filter(b,a,x),
subplot(3,1,1)
stem(n,y)
xlabel('n')
title('impulse response')

%Finding Step Response
x=zeros(1,length(n));
sr=find(n>=0);
x(sr)=1; %generating impulse sequence
y=filter(b,a,x),
subplot(3,1,2)
stem(n,y)
xlabel('n')
title('step response')

%Finding Sinusoidal Response
f=1/15;
x=0.5*sin(2*pi*f*n);
y=filter(b,a,x);
subplot(3,1,3)
stem(n,y)
xlabel('n')
```

```
title('sinusoidal response')
```

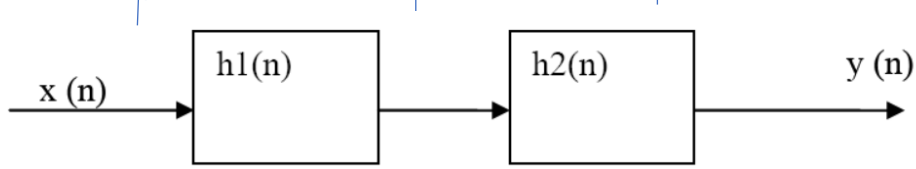


Activity List

Mark 10 **Time:** 45 minutes

Solve the following problems with MATLAB programming language

Question 1. Given x = [2, 3, 5, -2], h_1=[-1, 0, 3, 4] and h_2 = [2, 3 0, 7, 1]



Find **y(n).**

Question 2. Find the impulse response and step response of the following system [use filter() function]

y(n)= 0.6y(n-1)-0.08y(n-2)+x(n)

---END---